

Q.1 (5 marks)Find all possible values of  $i^{5/3}$ .Q.2 (5 marks)Find all the Fourier Series coefficients of the following  $T$ -periodic signal:

$$x(t) = e^{-t}, \quad 0 \leq t < T.$$

STOP at 9:18 am. (wall clock time)

Lecture - 11Recall:①. Mean-square convergence: ( $x$  is Riemann integrable)

$$\|x - S_N\|^2 \rightarrow 0 \quad \text{as } N \rightarrow \infty$$

②. Dirichlet-Jordan Test: (Pointwise convergence)If  $x$  is of bounded variation

$$S_N(t_0) \rightarrow \begin{cases} x(t_0) & \text{if } x \text{ is continuous at } t_0 \\ \frac{x(t_0^-) + x(t_0^+)}{2} \end{cases}$$

Special case of ②: Oppenheim, Wilsky &amp; Nawab.

↓

Dirichlet conditions.

I. Are these two types of convergence enough?

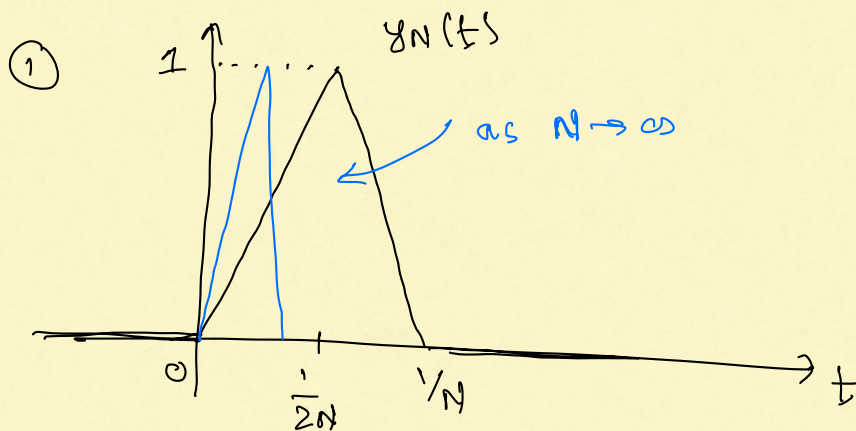
Examples:

Consider a sequence of functions  $y_1, y_2, \dots$

that converge pointwise to another function  $x$ .

$\Rightarrow$  for any  $t$ :

$$\lim_{N \rightarrow \infty} y_N(t) = x(t).$$



For  $t=0$ :  $y_N(t) = 0$

lim of the sequence:  $y_1(t), y_2(t), \dots$   
 $N \rightarrow \infty$

This is  $\lim_{N \rightarrow \infty} [0] = 0$

Now consider  $t > 0$ :

For  $N$  sufficiently large (in particular  $N > 1/t$ )

$$y_N(t) = 0$$

$$\Rightarrow \lim_{N \rightarrow \infty} y_N(t) = 0.$$



Thus,  $y_1, y_2, \dots$  converge pointwise to  $x$

where  $x(t) = 0$  for all  $t$ .

But note that for any  $N$  (no matter how large)

$$\max_t |y_N(t) - x(t)| = 1$$

$\Rightarrow$  The worst-case error between  $y_N$  and  $x$  does not vanish as  $N \rightarrow \infty$ .

Terminology:

The  $\infty$ -norm or supremum norm of a signal  $x$  is:

$$\|x\|_{\infty} = \sup_t |x(t)|$$

The worst-case error between  $x$  and  $y_N$  is:

$$\|x - y_N\|_{\infty} = \sup_t |x(t) - y_N(t)|$$

Definition: We say that  $y_1, y_2, \dots$  converges to  $x$  uniformly if:

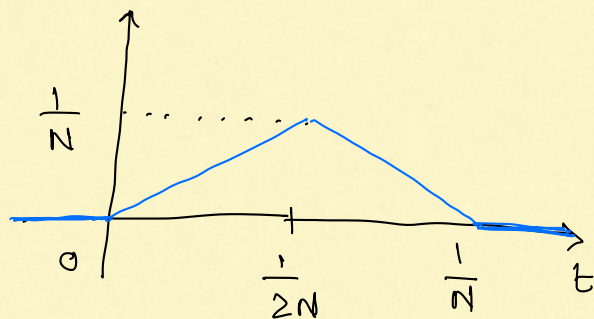
$$\|y_N - x\|_{\infty} \rightarrow 0 \text{ as } N \rightarrow \infty$$

$$\lim_{N \rightarrow \infty} \left[ \sup_t |y_N(t) - x(t)| \right] = 0$$

For previous example:  $\|y_N - x\|_{\infty} = \sup_t |x(t) - y_N(t)| = 1$

$\lim_{N \rightarrow \infty} (\|x - y_N\|_{\infty}) = \lim_{N \rightarrow \infty} (1) = 1 \Rightarrow$  Not uniform convergence.

Example ②: Define  $y_N(t)$



Let  $x(t) = 0 \quad \forall t$ .

$$\|x - y_N\|_{\infty} = \sup_t |x(t) - y_N(t)| = \frac{1}{N}$$

$$\lim_{N \rightarrow \infty} (\|x - y_N\|_{\infty}) = \lim_{N \rightarrow \infty} \left( \frac{1}{N} \right) = 0$$

$\therefore y_1, y_2, \dots \rightarrow x$  uniformly.

Notes:

① Pointwise convergence is weaker than uniform convergence

If  $y_1, y_2, \dots \rightarrow x$  pointwise:

then: there is no guarantee that  
 $y_1, y_2, \dots \rightarrow x$  uniformly.

But, if  $y_1, y_2, \dots \rightarrow x$  uniformly:

then:  $y_1, y_2, \dots \rightarrow x$  pointwise also.

In other words, if we want  $y_1, y_2, \dots \rightarrow x$  uniformly it means that we have a stronger requirement on how  $y_N$  behaves as  $N$  increases.



② Suppose  $y_1, y_2, \dots \rightarrow x$  uniformly.

That is:  $\|x - y_1\|_\infty, \|x - y_2\|_\infty, \|x - y_3\|_\infty, \dots \rightarrow 0$

this sequence of real numbers (sup. norms)  
converges to 0.

$\Rightarrow$  For any given  $\epsilon > 0$  there exists a  $N_\epsilon$  such that  
for all  $N > N_\epsilon$  we have  $\|x - y_N\|_\infty \leq \epsilon$ .

$\Rightarrow$  For any  $\epsilon > 0$  and large enough  $N$  we have:

$$\sup_t |x(t) - y_N(t)| \leq \epsilon$$

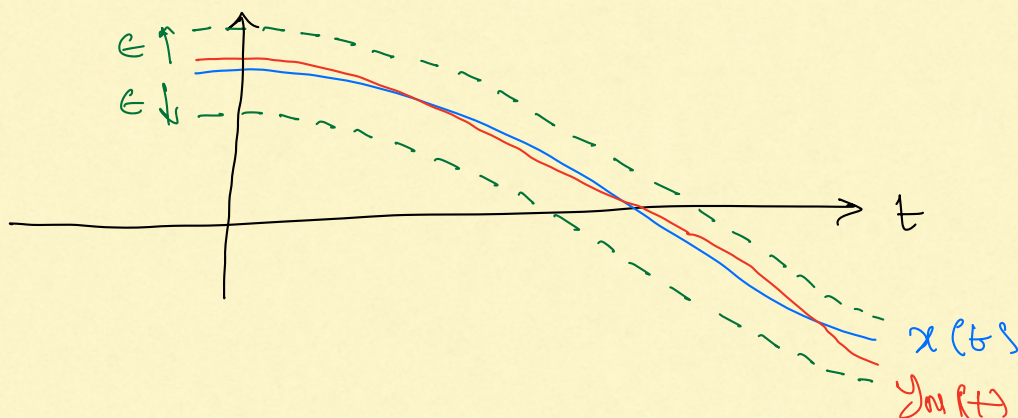
i.e.  $|x(t) - y_N(t)| \leq \epsilon$  for all  $t$ .

$\Rightarrow$  If we choose  $\epsilon > 0$  and  $N$  sufficiently large:

no matter which  $t$  we pick

$y_N(t)$  is only  $\epsilon$  away from  $x(t)$ .

Thus,  $y_N$  and  $x$  are close to each other  
across all time



Note: This does not happen with our  
first example in this lecture.